

DEPARTMENT OF ECONOMICS

MACROECONOMICS (ECN1215) JUNE 2017

TUTORIAL SHEET 6 SOLUTIONS

1. Define the following terms

a) Absolute advantage

The ability of one party to produce more of a good or service than competitors, using the least amount of resources.

b) Law of Absolute Advantage

When one country is more efficient than another country in the production of one commodity but is less efficient than the other nation in the production of a second commodity, then both nations can gain by each specializing in the production of the commodity of its absolute advantage and exchanging part of its output with the other nation for the commodity of its absolute disadvantage

c) Comparative Advantage

When a country can produce a good at a lower opportunity cost than another country.

d) Law of comparative advantage

Explains how mutually beneficial trade can take place even when one nation is less efficient than, or has an absolute disadvantage with respect to another nation in the production of all commodities. The less efficient nation should specialize in and export the commodity in which its absolute disadvantage is smaller (the commodity of its comparative advantage) should import the other commodity.

2. Trade policy usually involve trade restrictions such as tariffs, quotas and voluntary exports restraints (VERs). What are tariffs, quotas and VERs?

a) **Voluntary Exports Restraints (VERs)**: Refers to the case where an importing country induces another nation to reduce its exports of a commodity voluntarily under the threat of higher trade restrictions when their exports threaten an entire domestic industry.

- b) **Tariff**: A tax or duty levied on an imported or exported commodity when it enters or leaves the country. I.e. it can either be an export tariff or import tariff.
- c) **Quota**: is a restriction specifying the maximum amount of a good that may enter a country during a specified time period.

3. Explain the following arguments for trade restrictions:

a) **Infant industry argument**

Protection to allow industries to reach the size and scale that is economically viable in the international competitive world.

[**Note**: if a question just requires a definition, then the above sentence is enough. But if you're required to explain, then add this part: A nation may have a potential comparative advantage in a commodity but because of lack of know-how and initial small level of output, the industry will not be set up or, if already started, cannot compete successfully with more established foreign firms. Therefore, temporary protection is justified to establish and protect the domestic industry during its infancy until it can meet foreign competition, achieve economies of scale, and reflect the nation's long-run comparative advantage. After achieving this, protection should be removed]

b) **Dumping**

The illegal practice of selling a good in a foreign market at a lower price than prevails in the market of the exporting nation with the intent to drive the foreign company out of the market and then raise the prices.

c) **Countervailing duties**

Entails imposing tariffs (by the importing countries) on imported goods that are produced by firms that receive production subsidies from their governments.

4. Define an exchange rate and differentiate between a flexible exchange rate system and a fixed exchange rate system.

Exchange rate is the price for which the currency of a country can be exchanged for another country's currency.

Flexible exchange rate system: is one in which the exchange rate is determined by market forces of demand and supply. Zambia uses this system, but once in a while the Bank of Zambia intervenes by selling reserves to restore the kwacha if it depreciates too much.

Fixed exchange rate system: is where the exchange rate is pre-determined and set by the government at a particular level. It requires the government to buy or sell sufficient quantities of their currencies to keep the change rate at an agreed value. E.g China uses this system.

[Note: You should also know the difference between nominal and real exchange rates, use your handouts]

5. What is the Balance of Payments (BOP)? Explain the 3 components of the BOP?

The BOP is a record of all transactions made between one particular country and all other countries during a specified period of time.

Components of the BOP include:

- i. **Current account or Trade account:** records all exports and imports of goods and services, and all other items that do not result in addition or subtraction on the country's claim on foreign resources.
 - ii. **Capital account:** records all transactions that affect the amount of claims that a country has abroad and that other countries have in this country.
 - iii. **Official settlement account:** shows the change in a country's holding of foreign currency
6. You are given the information shown in the table about the production relationship between Portugal and England:

	Labor Hours per Bottle of Wine	Labor Hours per Pound of Cheese
Portugal	10	5
England	6	2

- a) Which country has absolute advantage in wine? In cheese?
- b) Which country has comparative advantage in wine? In cheese?

- a. England has absolute advantage on producing both wine and cheese because its labor requirements are lower. To produce 1 bottle wine Portugal requires 10 hours but England only requires 6 hours. To produce 1 pound of cheese Portugal requires 5 hours but England only requires 2 hours.

- b. *Comparative advantage means lower opportunity costs.*

	<i>Opportunity cost</i>	
	Wine	Cheese
Portugal	$\frac{10}{5} = 2$	$\frac{5}{10} = 0.5$
England	$\frac{6}{2} = 3$	$\frac{2}{6} = 1.33$

Therefore, Portugal has comparative advantage in producing wine and England has comparative advantage in producing cheese.

7. The following table gives recent figures for yield per acre in Illinois and Kansas:

	Wheat	Soyabeans
Illinois	48	39
Kansas	40	24

- If we assume that farmers in Illinois and Kansas use the same amount of labor, capital, and fertilizer, which state has an absolute advantage in wheat production? Soyabeans production?
- If we transfer land out of wheat into soybeans, how many bushels of wheat do we give up in Illinois per additional bushel of soybeans produced? In Kansas?
- Which state has a comparative advantage in wheat production? In soyabeans production?

- a) **Illinois has an absolute advantage in both wheat and soyabeans**

- b) **In Illinois, taking 1 acre out of wheat and moving it into soyabeans sacrifices 48 bushel for 39 bushels of soyabeans. This is $\frac{48}{39} = 1.23$ bushels of wheat for each bushel of soyabeans. In Kansas, the sacrifice is $\frac{40}{24} = 1.67$ bushels of wheat for each bushel of soyabeans.**

c) **Comparative advantage means lower opportunity costs.**

	Opportunity cost	
	Wheat	soyabeans
Illinois	$\frac{39}{48} = 0.81$	$\frac{48}{39} = 1.23$
Kansas	$\frac{24}{40} = 0.6$	$\frac{40}{24} = 1.67$

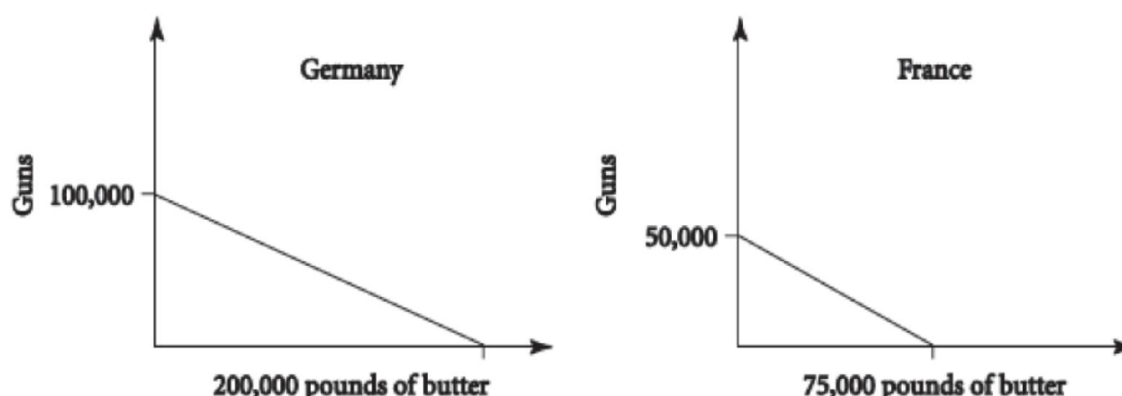
Kansas has a comparative advantage in wheat because it has a lower opportunity cost in wheat production. Illinois has a comparative advantage in soyabean because it has a lower opportunity cost in soyabean production.

8. Suppose Germany and France each produce only two goods, guns and butter. Both are produced using labor alone. Assuming both countries are at full employment, you are given the following information:

Germany: 10 units of labor required to produce 1 gun 5 units of labor required to produce 1 pound of butter Total labor force: 1,000,000 units

France: 15 units of labor required to produce 1 gun 10 units of labor required to produce 1 pound of butter Total labor force: 750,000 units

- a) Draw the production possibility frontiers for each country in the absence of trade.



- b) If transportation costs are ignored and trade is allowed, will France and Germany engage in trade? Explain.

Yes. The opportunity cost of a gun in Germany is 2 pounds of butter. The opportunity cost of a gun in France is only 1.5 pounds of butter. France has a comparative advantage in guns. Similarly, the opportunity cost of a pound of butter in Germany is 1/2 of a gun, whereas the opportunity cost of a pound of butter in France is 2/3 of a gun. Germany has a comparative advantage in the production of butter.

- c) If a trade agreement is negotiated, at what rate (number of guns per unit of butter) would they agree to exchange?

As long as the agreement specifies between 1.5 pounds of butter and 2 pounds of butter per gun, specialization and trade will benefit both countries. For example, an agreement to exchange 1.75 pounds of butter per gun would benefit both countries

9. The United States and Russia each produce only bearskin caps and wheat. Domestic prices are given in the following table:

	RUSSIA	USA	
Bearskin caps	10 Ru	\$7	Per hat
Wheat	15 Ru	\$10	Per bushel

Suppose the exchange rate is \$1 = 1 Ru.

- a) Which country has an absolute advantage in the production of bearskin caps? Wheat?

You cannot tell from the information given which country has an absolute advantage because you are not given any information that would indicate the actual quantities of inputs used in production in either country. You cannot tell from the information given which country has an absolute advantage because you are not given any information that would indicate the actual quantities of inputs used in production in either country.

- b) Which country has a comparative advantage in the production of bearskin caps? Wheat?

If resources are fully mobile between sectors, the opportunity cost of a cap is 2/3 of a bushel of wheat in Russia; the opportunity cost of a cap is 7/10 of a bushel of wheat in the United States. Russia has a comparative advantage in cap production. The opportunity cost of a bushel of wheat in Russia is 1.5 caps. The opportunity cost of a bushel of wheat in the United States is 10/7, or 1.43, caps. The United States has a comparative advantage in wheat

- c) If the United States and Russia were the only two countries engaging in trade, what adjustments would you predict assuming exchange rates are freely determined by the laws of supply and demand?

At \$1 = 1 Ru, both goods in the United States are cheaper to everyone. That means that there was a big demand for dollars and no supply on foreign exchange markets. The price of the dollar would rise. When a dollar was valued at between 1.43 Ru (calculated as 10 Ru/\$7) and 1.50 Ru (calculated as 15 Ru/\$10), caps would be cheaper in Russia and wheat would be cheaper in the United States. If the price of a dollar rises to more than 1.50 Ru, everyone would buy both goods in Russia.

10. Australia and the United States produce white and red wines. Current domestic prices for each wine are given in the following table:

	AUSTRALIA	USA
White wine	5 AU\$	10 US\$
Red wine	10 AU\$	15 US\$

Suppose the exchange rate is 1 AU\$ = 1 US\$.

- a) If the price ratios within each country reflect resource use, which country has a comparative advantage in the production of red wine? White wine?

The opportunity cost of a bottle of red wine is 1.5 bottles of white in the United States and 2 bottles of white in Australia. The United States, therefore, has a comparative advantage in red wine. The opportunity cost of a bottle of white wine is .66 bottles of red in the United States and .5 bottle of red in Australia. Australia, therefore, has a comparative advantage in white wine.

- b) Assume that there are no other trading partners and that the only motive for holding foreign currency is to buy foreign goods. Will the current exchange rate lead to trade flows in both directions between the two countries? Explain.

No, at the current exchange rate, both white and red wine are cheaper in Australia. U.S. citizens will want to import both types of wine from Australia, but Australians will not want to import U.S. wine.

- c) What adjustments might you expect in the exchange rate? Be specific.

In this situation, we would expect the price of the dollar to decrease until U.S. red wine became attractive to Australians whereas Australian white wine remained attractive to Americans. [An

exchange rate between 1.5 (calculated as 15 US\$/10 AU\$) and 2 (calculated as 10 US\$/5 AU\$) U.S. dollars to 1 Australian dollar would accomplish this.]

- d) What would you predict about trade flows between Australia and the United States after the exchange rate has adjusted?

In the long run, we would expect exchange rates to adjust until Americans were exporting red wine to Australia and Australians were exporting white wine to the United States.